

GEOG 515 Final Project Reflection

I began this project with a basic interest in investigating walkability. It's a topic I have always had some interest in; a personal draw to this university for me was the level of walkability around campus, and I have always thought of how sharp that walkability falls once off campus grounds. Just seeing that distribution on the map was my initial hook into this project

My first task was to locate data. It didn't take long for me to find the national walkability index, part of the EPA Smart Location Database. I knew I wanted to limit my scope to something locally impactful, but large enough to procure data, which is how I landed on the idea of the NC Triangle. Initially, I wasn't sure exactly where to mark my extent. I ended up getting the idea from exercise 13 in class, where we defined the triangle region around the five counties of Wake, Durham, Orange, Chatham, and Alamance.

When it came to moving my project forward, the first place my mind went to with walkability was a social justice angle. When looking into income data available at a block group level, the easiest source was the percentage of low- and moderate-income residents. Once I had this variable, I had a good vision of where I wanted to go with the project. I have previous experience using regression analysis and controlling for spatial autocorrelation in GeoDa, so moving that into R felt like a natural step for me.

The biggest issue that got in the way was when I realized my variables were mismatched (GEOID10 vs GEOID20). Thankfully, I have encountered this before and knew there were a few ways I could approach it. I decided on a relatively easy method: aggregating GEOID20 to match GEOID10 and weighing the variable by area taken up. If I was doing this again, I might've decided on a different method, like weighing on population.

I found the suggestion under Project Task 2 to expand the project probably the most helpful piece of feedback I received. It inspired me to include population density as a factor within the model, the idea being that I wanted to control for that before considering the relationship with income.

Completing the project gave me more experience working with spatial data and statistical models within R. I was lucky to find some of our final exercises in class line up with the kind of work I was hoping to do with this project; it helped me both improve the project and gave me more fundamentals to consider. I also feel like I learned a lot about the importance of keeping a project organized. That idea of handing off your project to your classmate in PT4, for example, was a really strong reminder of why we do the things we do. Every piece of file structure, naming convention, etc. will be viewed by somebody else trying to understand your work.